

Body-fitted finite-volume DFG 2D-1 cylinder verification

Actual body-fitted Gmsh meshes; strict physical relative error <3%.

Scheme: {'momentum_convection': 'linear-upwind', 'pressure_gradient': 'gauss-linear', 'face_interpolation': 'projected-linear', 'wall_gradients': 'linear'}

CPU calculations; no new GPU or matched TensorLBM result.

Positive mesh geometry and nonorthogonality gate precede all solves.

scale-2: 2178 cells; converged=True; accepted=False

{'drag': 5.33545599723707, 'lift': 0.02458746467726144, 'pressure_difference_pa': 0.11064068123044879}

{'drag_relative_error': 0.0437454, 'lift_relative_error': 1.3154332, 'pressure_difference_pa_relative_error': 0.0585388}

scale-1: 8252 cells; converged=True; accepted=False

{'drag': 5.5126498036744165, 'lift': 0.012031717391097858, 'pressure_difference_pa': 0.11517945530273684}

{'drag_relative_error': 0.0119876, 'lift_relative_error': 0.1330423, 'pressure_difference_pa_relative_error': 0.0199175}

scale-0.5: 32274 cells; converged=True; accepted=True

{'drag': 5.558997597015046, 'lift': 0.010385328610916589, 'pressure_difference_pa': 0.11648447002326939}

{'drag_relative_error': 0.0036809, 'lift_relative_error': 0.0220003, 'pressure_difference_pa_relative_error': 0.0088129}

Full independently replayed equations: True

Refinement sequence accepted: False

Asymptotic grid regime qualified: False

Problem, numerical method and reproducibility

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Problem and reference

DFG 2D-1: steady incompressible laminar flow, $Re=20$, channel 2.2 by 0.41 m, cylinder center (0.2,0.2) m and radius 0.05 m. The mean inlet speed is 0.2 m/s, density 1 kg/m³ and dynamic viscosity 0.001 Pa s. No-slip applies to the cylinder and channel walls. The inlet parabolic velocity is integrated exactly over each physical face; outlet pressure is zero with zero-gradient velocity. Drag, lift and front/rear pressure difference are compared with the [FeatFlow reference](https://wwwold.mathematik.tu-dortmund.de/~featflow/en/benchmarks/cfdbenchmarking/flow/dfg_benchmark1_re20.html).

Mesh and discretization

Gmsh 4.15.2 frontal Delaunay generates conforming triangles with physical boundary tags; cylinder CAD arcs or a closed NACA0012 interpolating spline define the body. Physical wall faces are linear chords. See the [Gmsh official manual](<https://gmsh.info/doc/texinfo/gmsh.html>) for algorithm 6 and Distance/Threshold fields. Neither Cartesian obstacle masks nor staircase boundaries are used. Positive volumes and projected face distances and a maximum nonorthogonality below 70 degrees are required before solving. The 70-degree project gate is a screening criterion, not proof of sufficient accuracy.

The recorded discretization is ``{"face_interpolation": "projected-linear", "momentum_convection": "linear-upwind", "pressure_gradient": "gauss-linear", "wall_gradient": "incompressible-normal"}``. Linear upwind uses the least-squares gradient of the upstream cell and the actual face-to-cell displacement; its conservative deferred correction is inspired by [OpenFOAM linearUpwind](<https://github.com/OpenFOAM/OpenFOAM-dev/blob/master/src/finiteVolume/interpolation/surfaceInterpolation/schemes/linearUpwind/linearUpwind.C>). This is unlimited reconstruction, so no bounded/TVD claim is made. Gauss pressure gradients use geometric face interpolation and the same boundary face pressure used by pressure traction. Nonorthogonal diffusion, physical Rhie-Chow flux and no-slip reconstructed molecular stress are retained. Cylinder acceleration uses a full-grid coupled Picard matrix with an independent response check; NACA uses SIMPLE and Anderson mixing. CPU sparse solves check true residuals. No reference value enters the equations, initialization or relaxation.

Wall-traction comparison

The numerical solution is reused unchanged from verification-gmsh-cylinder. This report changes only viscous wall-traction postprocessing; cell velocity, pressure, mass flux, steady histories and pressure traction are bitwise identical to the archived input. The source field SHA256 and original numerical producer snapshot are supplied. At a stationary incompressible no-slip wall, tangential velocity derivatives vanish and the normal derivative of normal velocity is zero. The reconstruction therefore estimates only the normal derivative of tangential velocity from its owner-cell value and normal distance. It is a first-order wall-shear estimate; the finite cell-to-wall reconstruction remainder contains higher-order normal velocity and is not the prescribed wall velocity. A separate divergence-free circular swirl with exact wall shear gives relative L2 errors of 5.005%, 2.508% and 1.342% on these meshes; this analytic reconstruction test does not replace the solved external-flow fields. Original general-gradient traction results remain archived.

Reproduction: ``PYTHONPATH=src python scripts/postprocess_incompressible_wall.py --input results/cylinder --output results/cylinder-wall`` after generating the original cylinder study. The commands below reproduce the upstream flow solve.

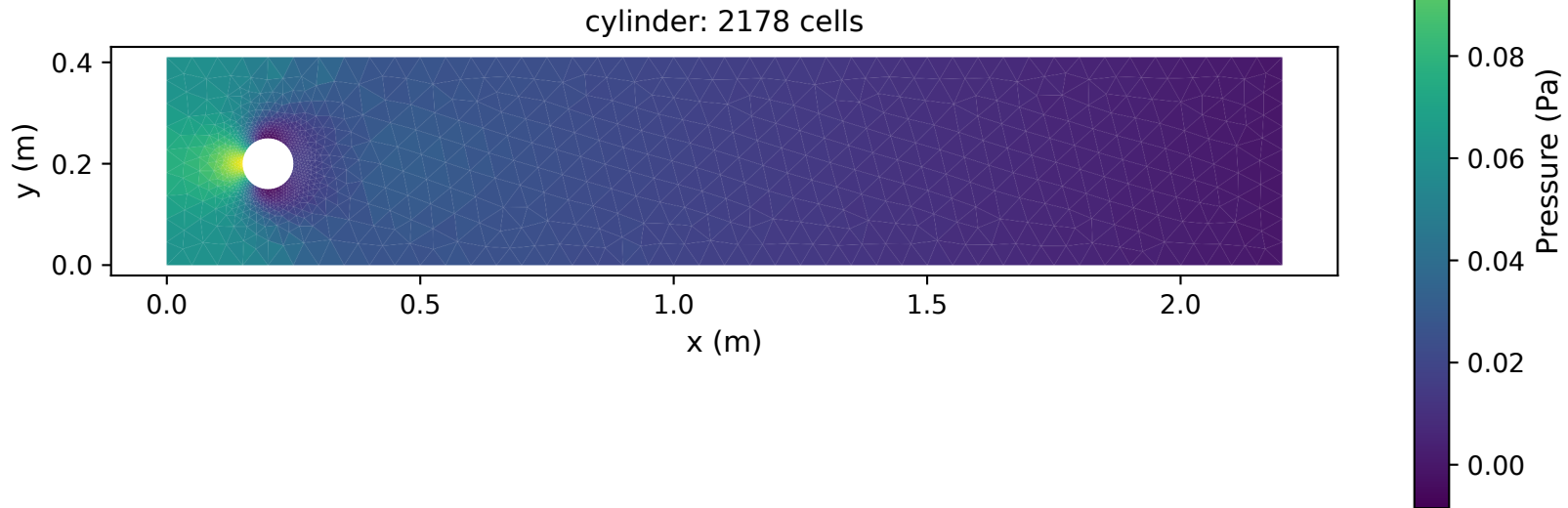
Reproduction and acceptance

```
```bash
uv pip install --python /path/to/python gmsh numpy scipy matplotlib
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 PYTHONPATH=src python -m
tensorfvm.verification.external_gmsh --case cylinder --scales 2.0 1.0 0.5 --solver conservative-
linear-upwind --max-iterations 300 --output results/cylinder
PYTHONPATH=src python scripts/audit_external_equations.py results/cylinder
PYTHONPATH=src python -m tensorfvm.verification.external_mesh_report results/cylinder
```
```

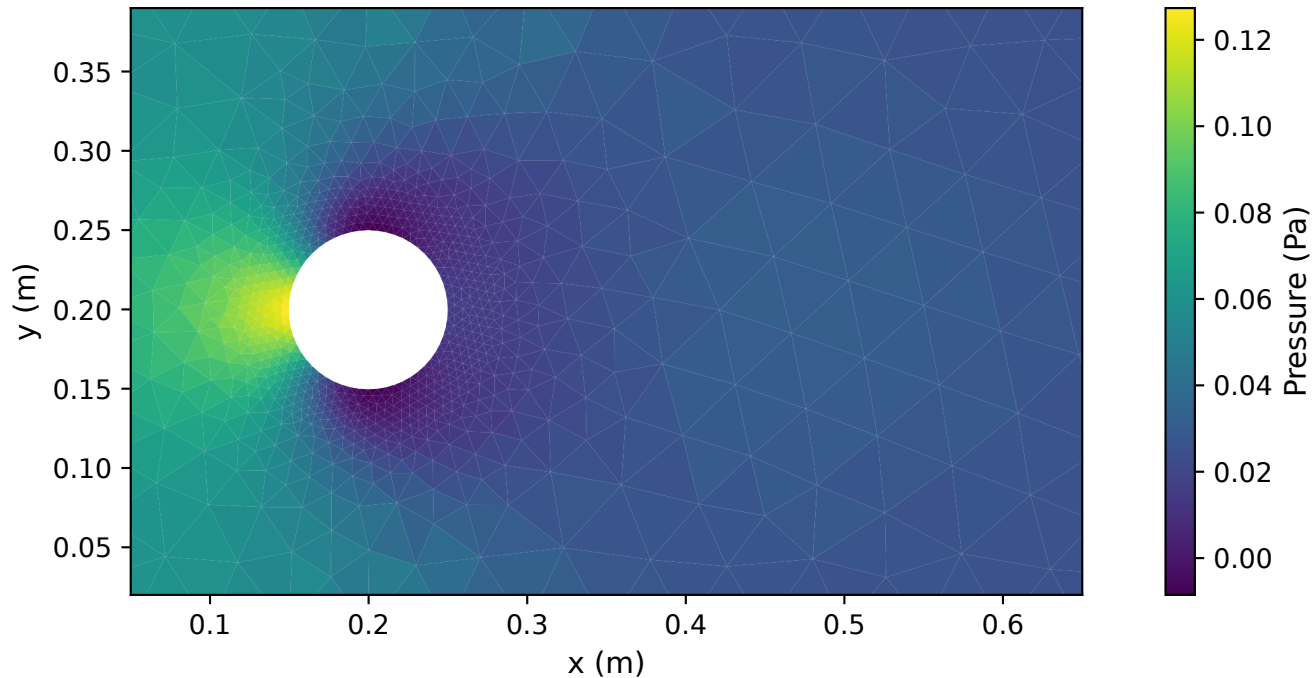
Raw mesh/field files, velocity/pressure arrows, face mass fluxes, separate tractions, full

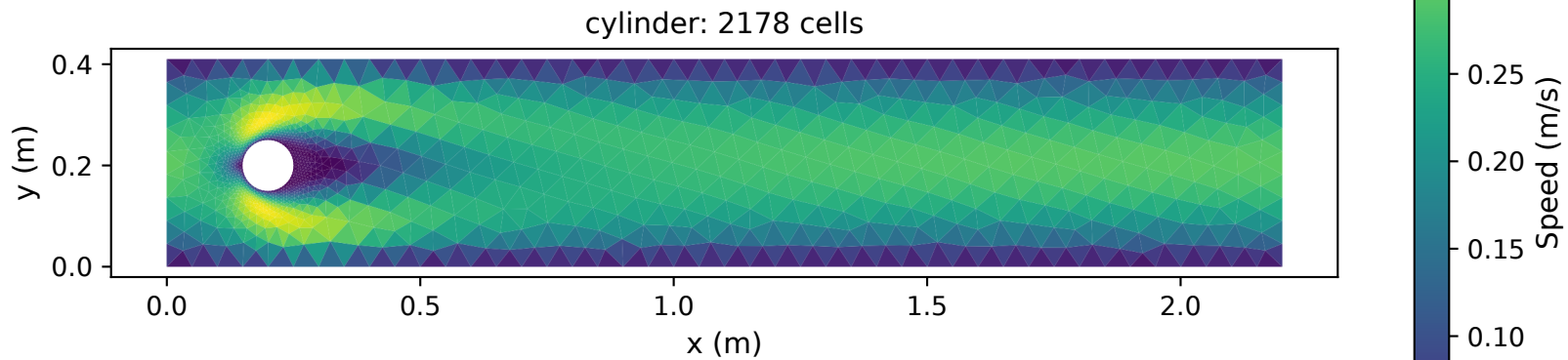
Problem, numerical method and reproducibility

histories, linear residuals and SHA256 source/artifact manifests accompany this report. Each physical relative error must be strictly below 3%; steady equation residuals must be below $1e-6$ and the physical flux fixed-point defect below $1e-7$. Source-preserving independent NumPy replay checks full configured equations. Three runs do not establish an asymptotic grid regime; qualification of the finest run does not qualify the entire refinement sequence. GPU and matched TensorLBM runs have not been performed for this new unstructured path.

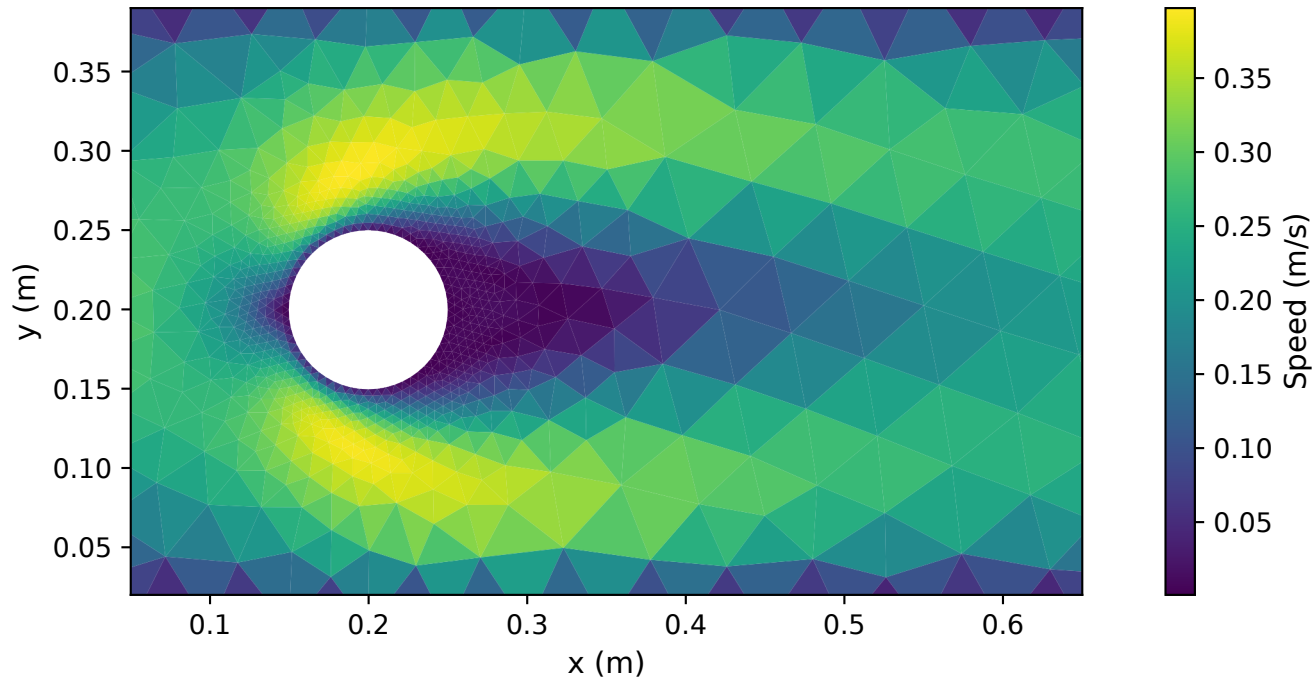


cylinder: 2178 cells

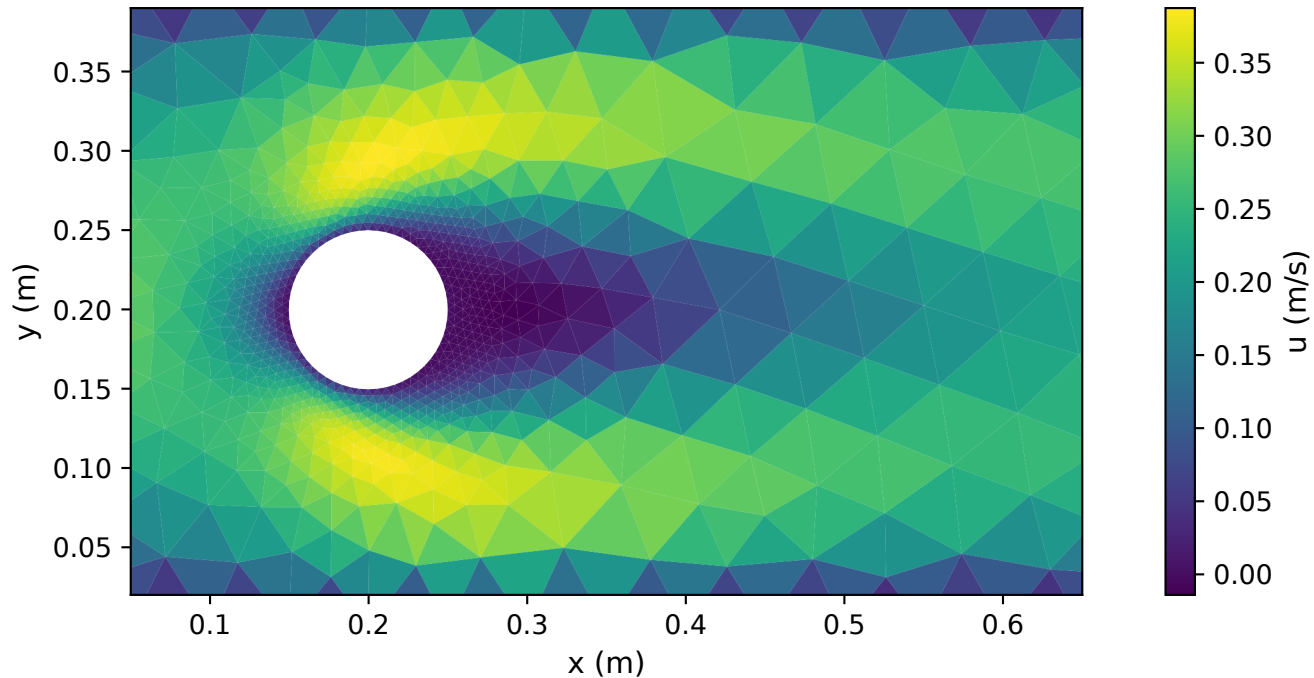




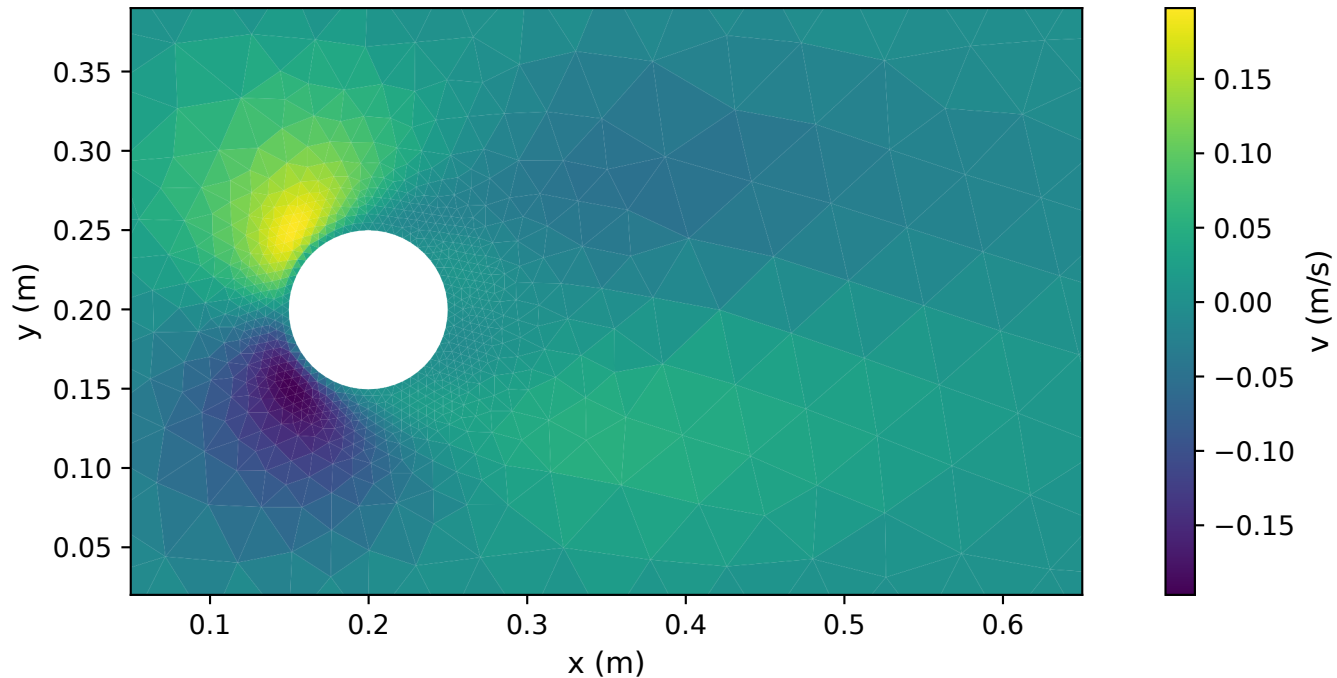
cylinder: 2178 cells



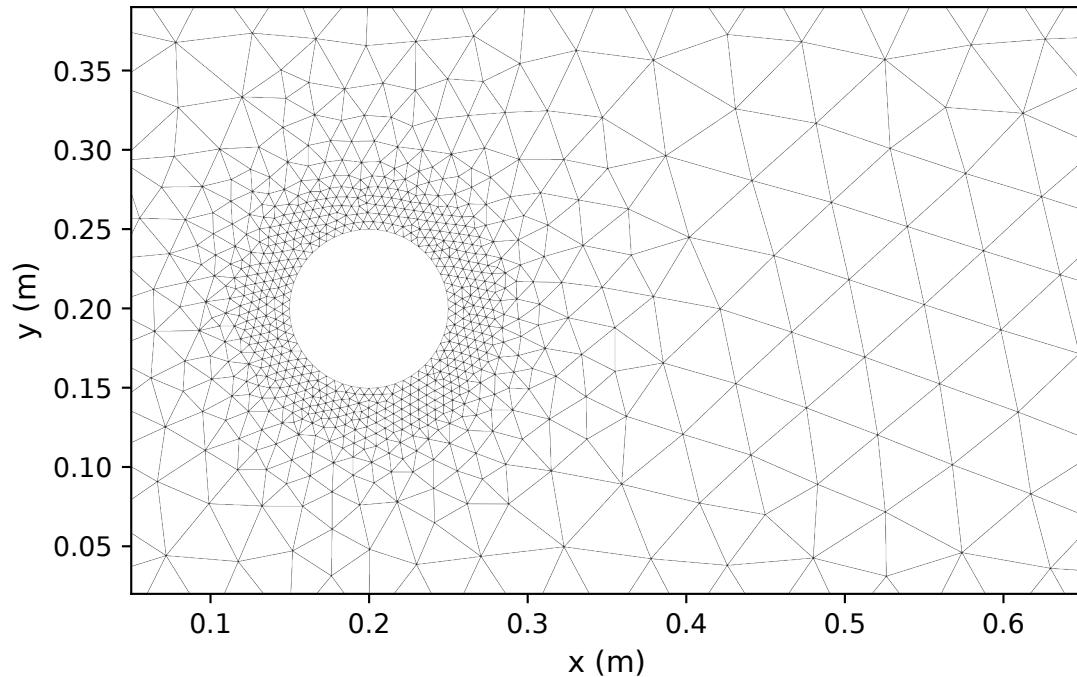
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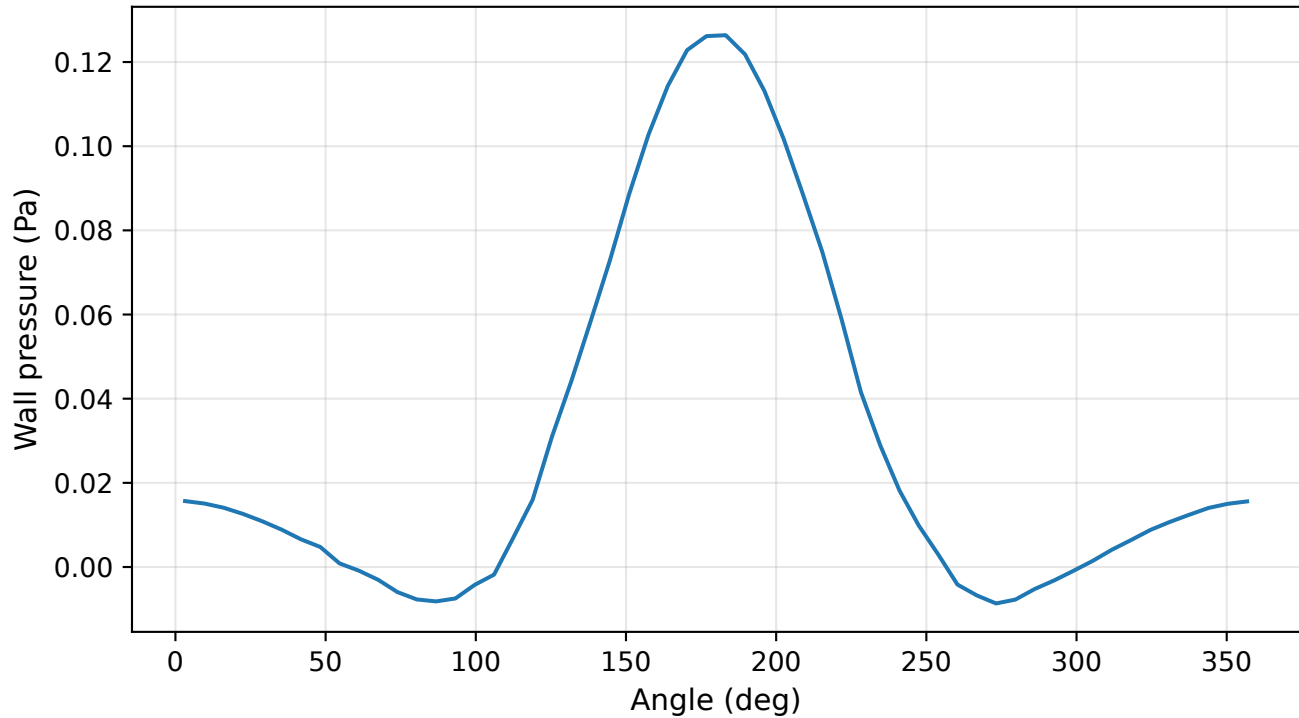


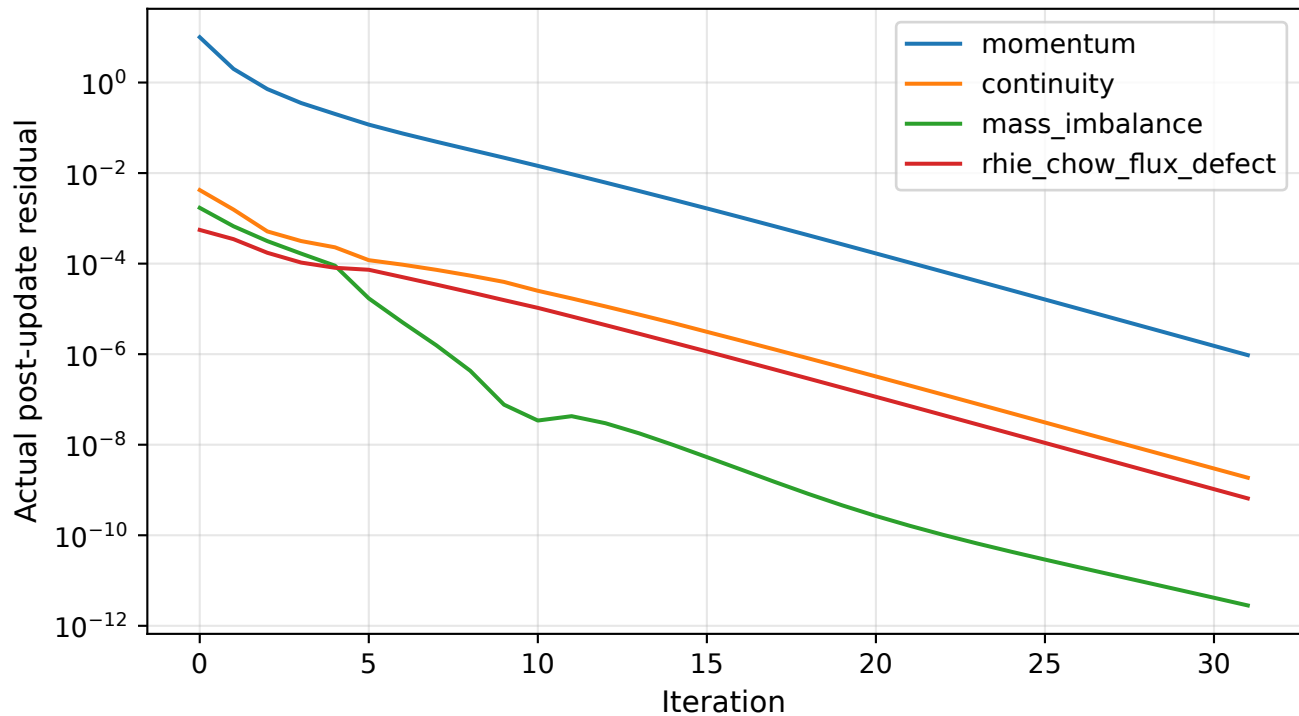
cylinder: 2178 cells

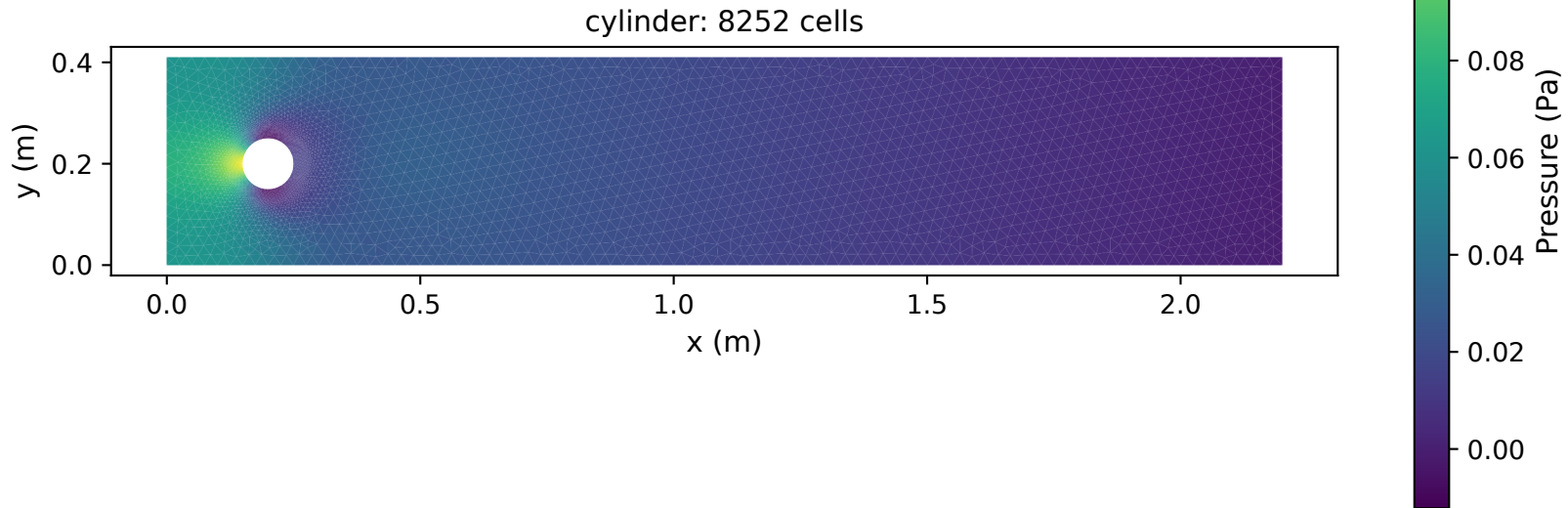


Actual body-fitted mesh

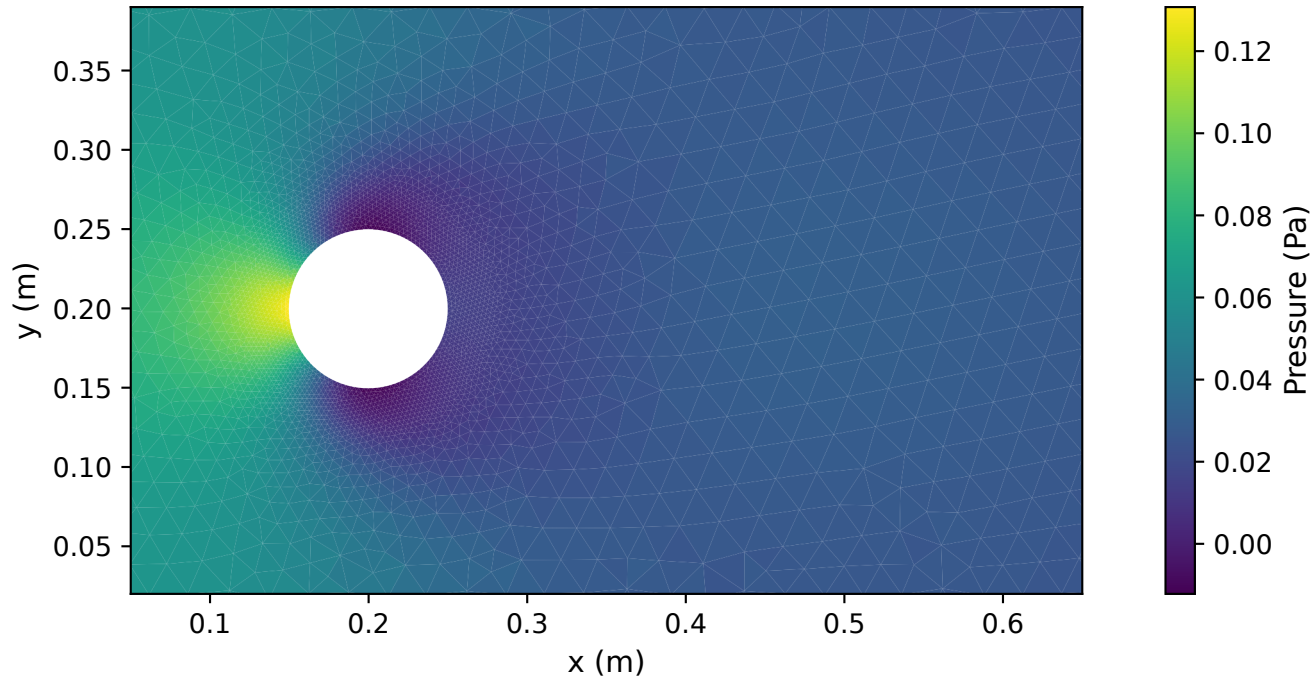


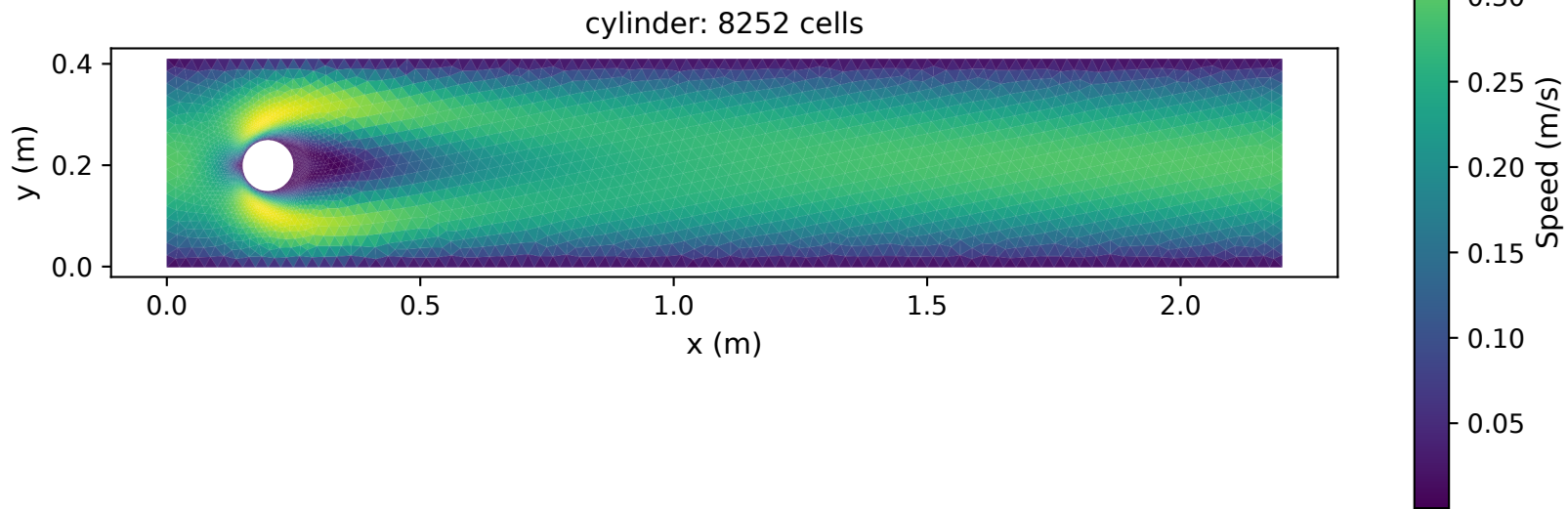




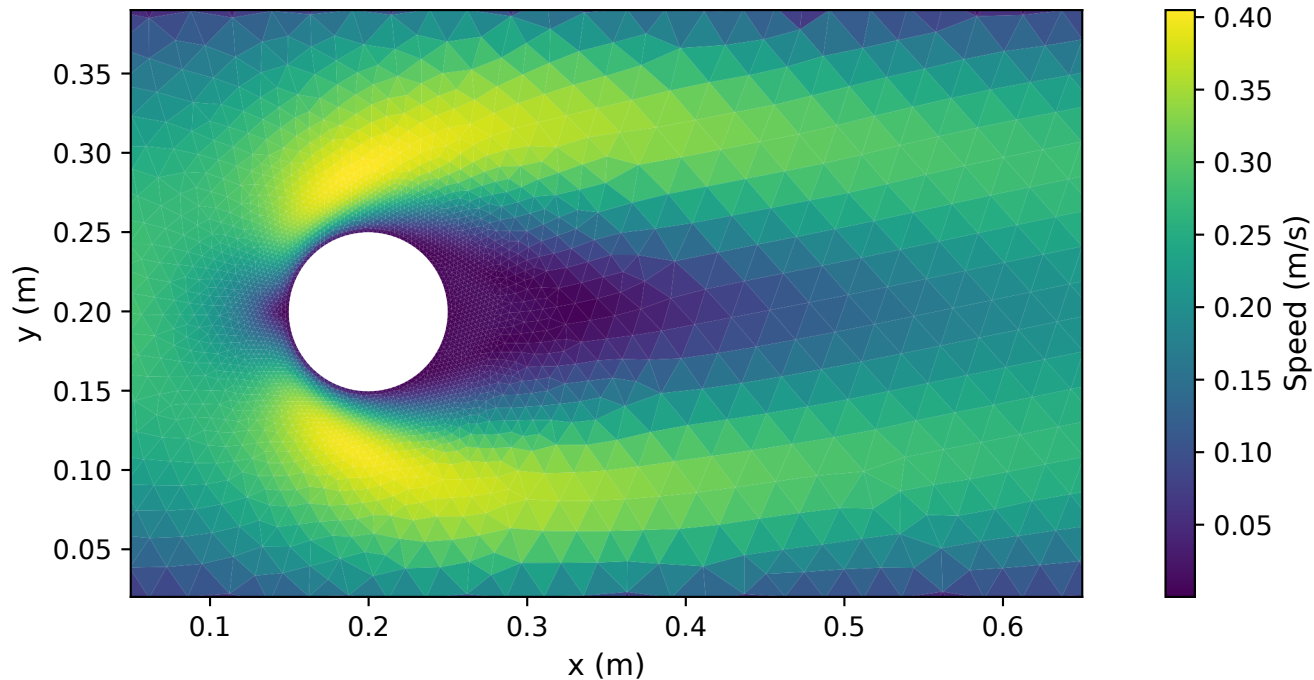


cylinder: 8252 cells

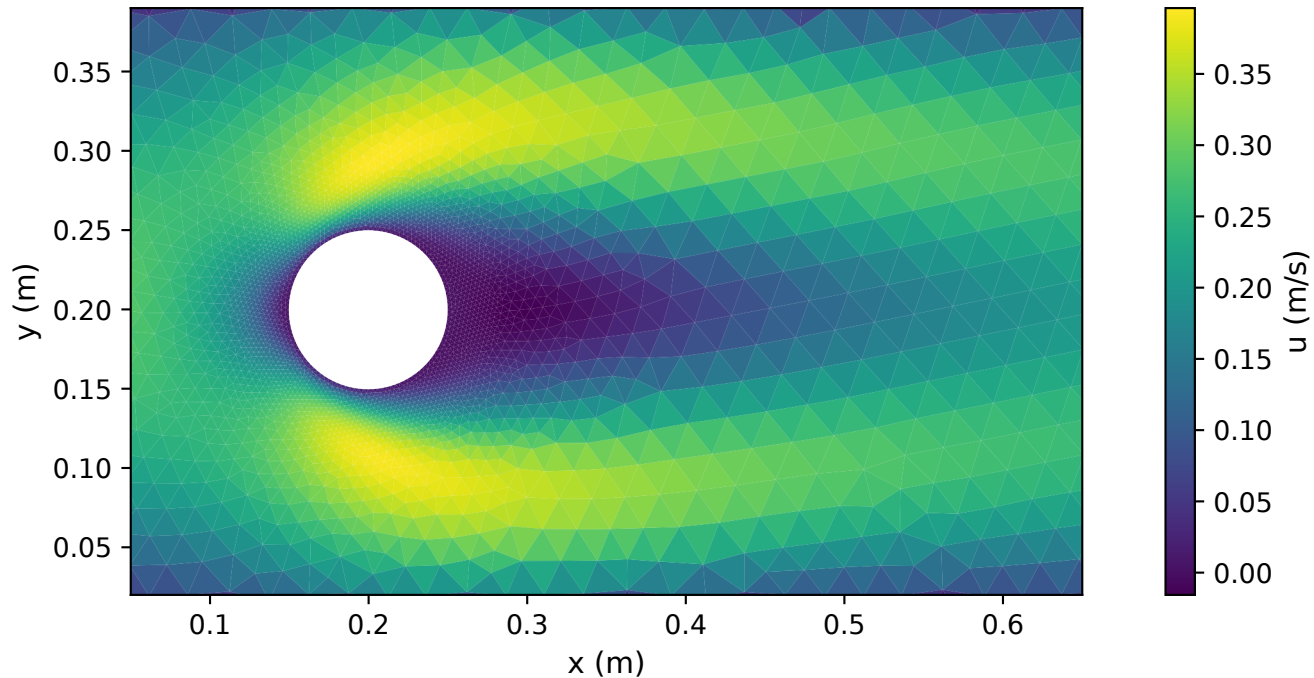




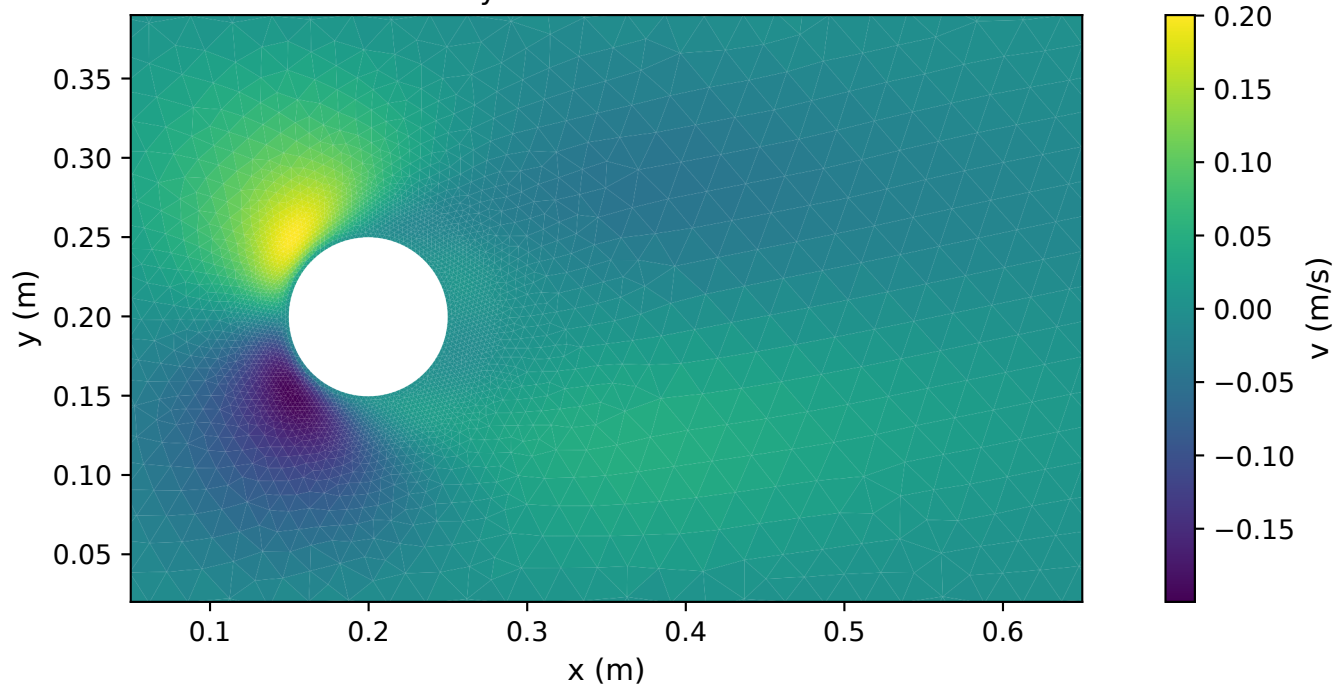
cylinder: 8252 cells



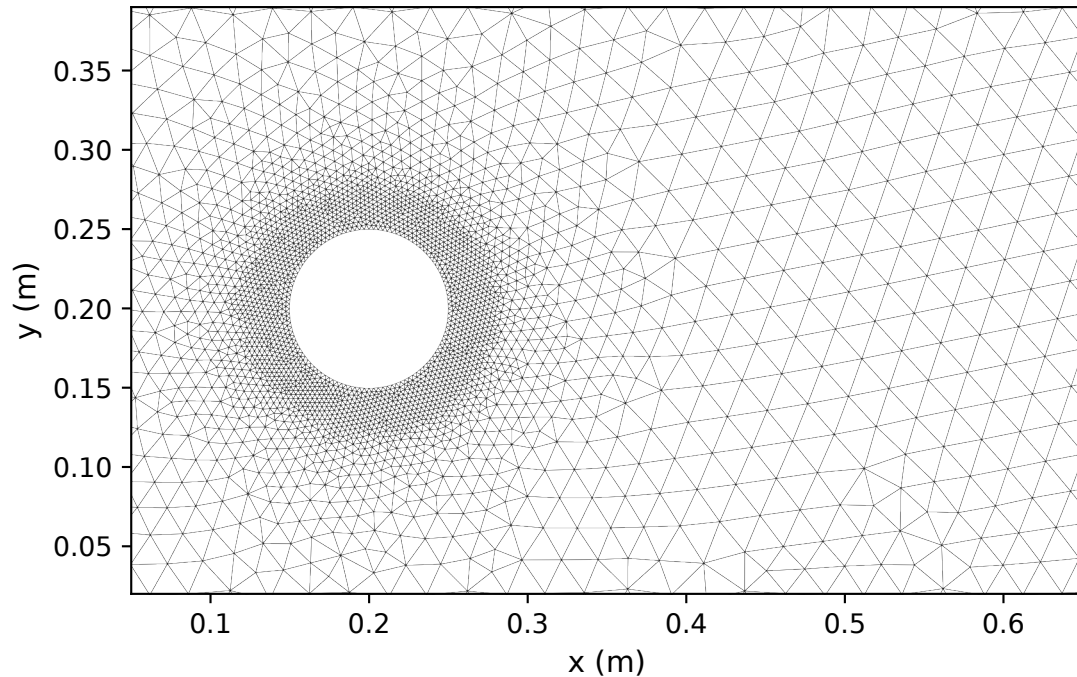
cylinder: 8252 cells

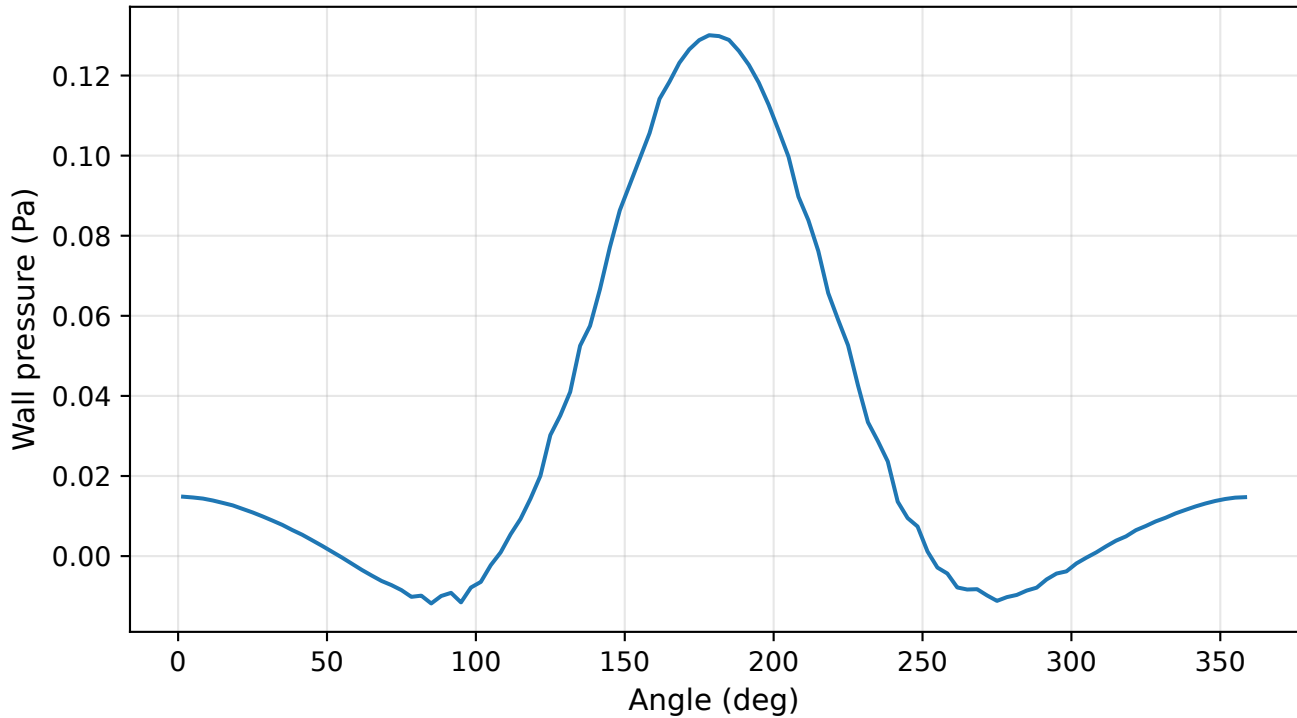


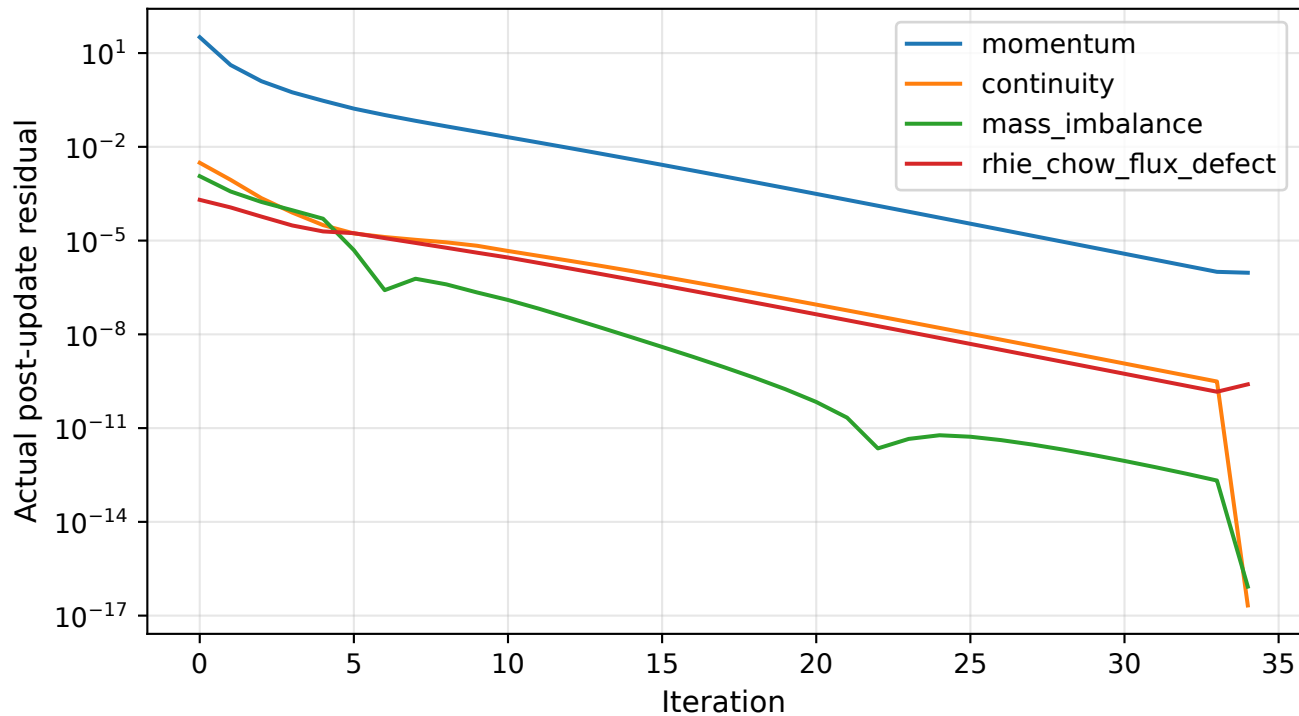
cylinder: 8252 cells

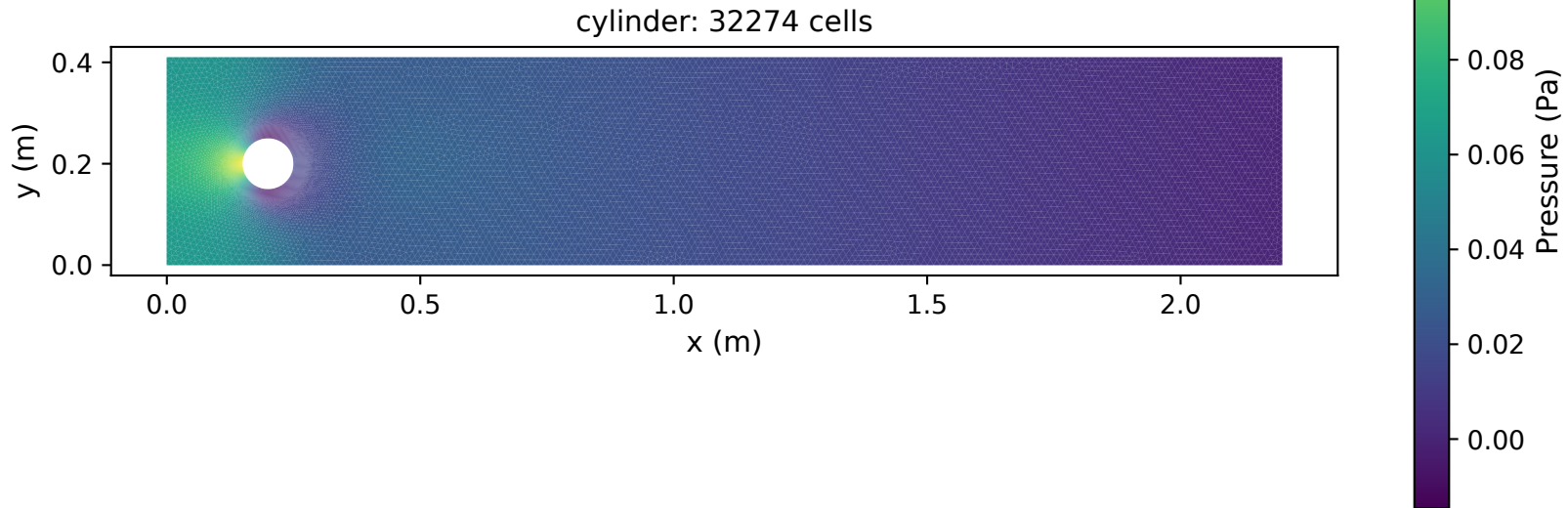


Actual body-fitted mesh

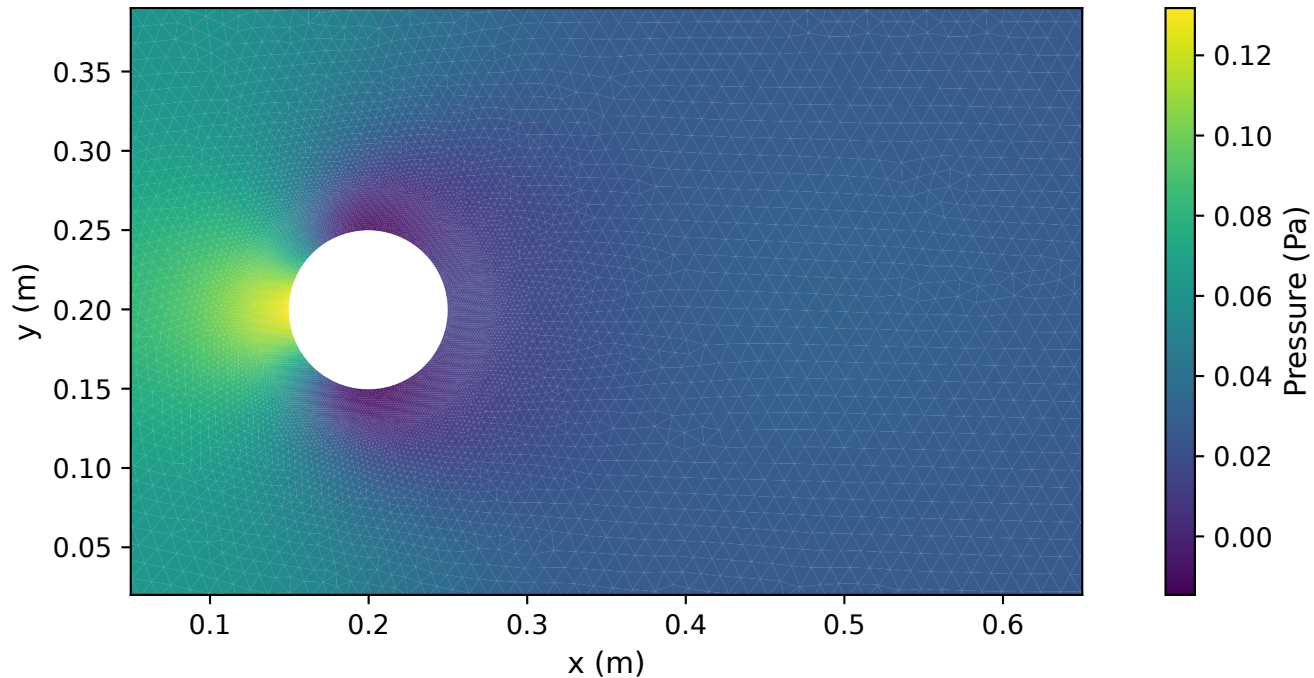


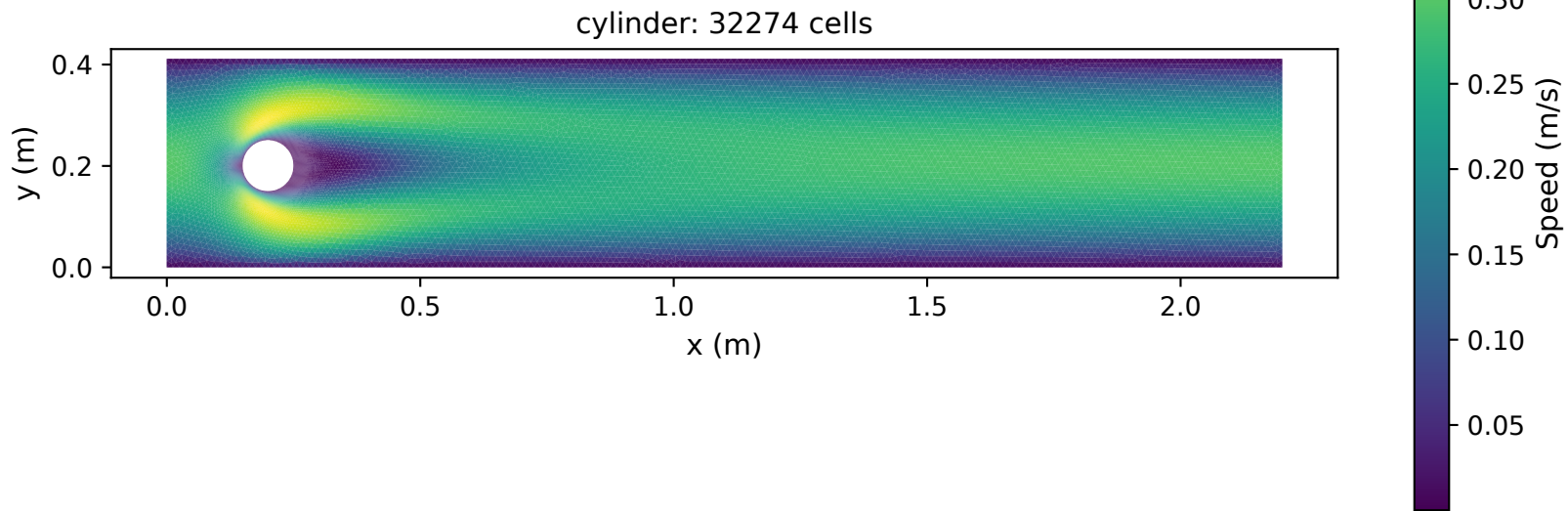




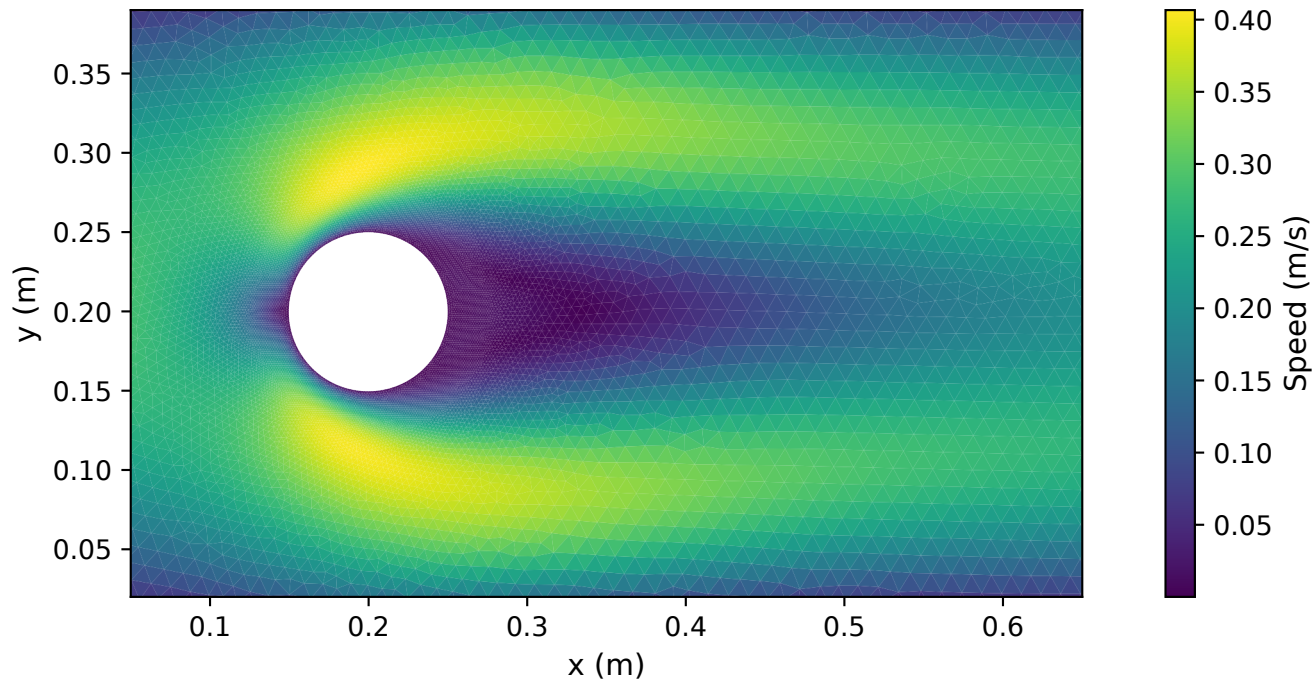


cylinder: 32274 cells

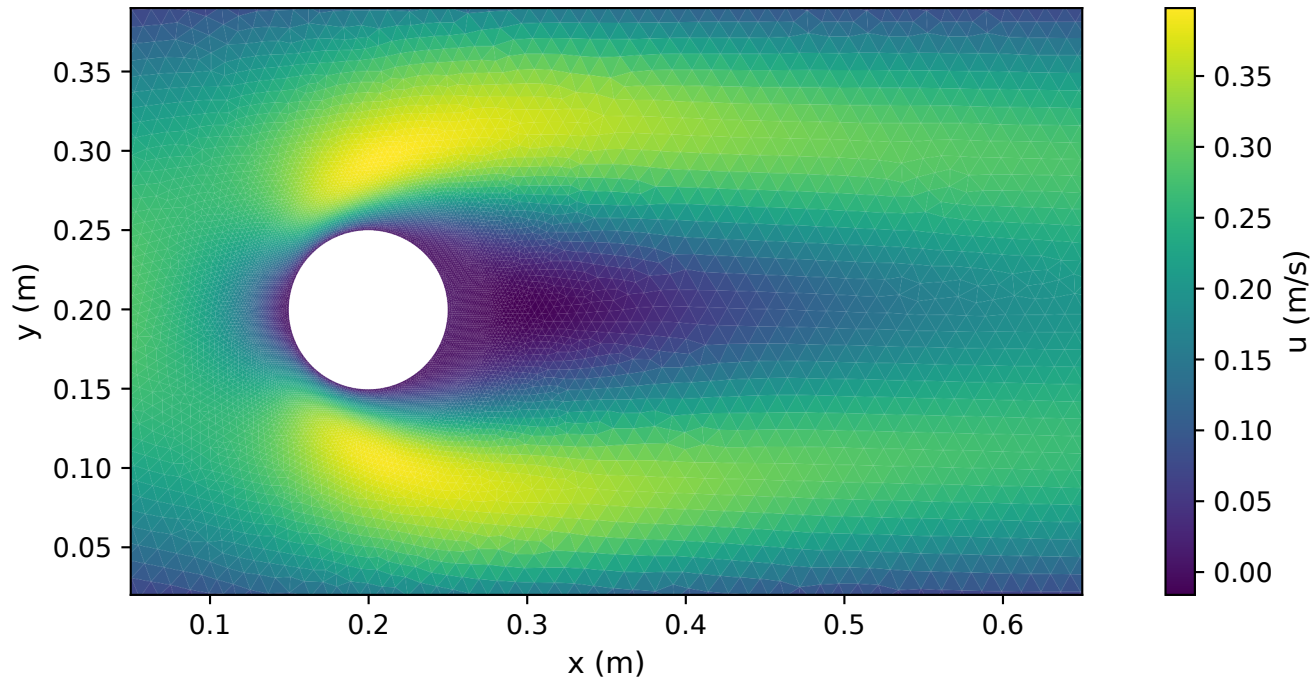




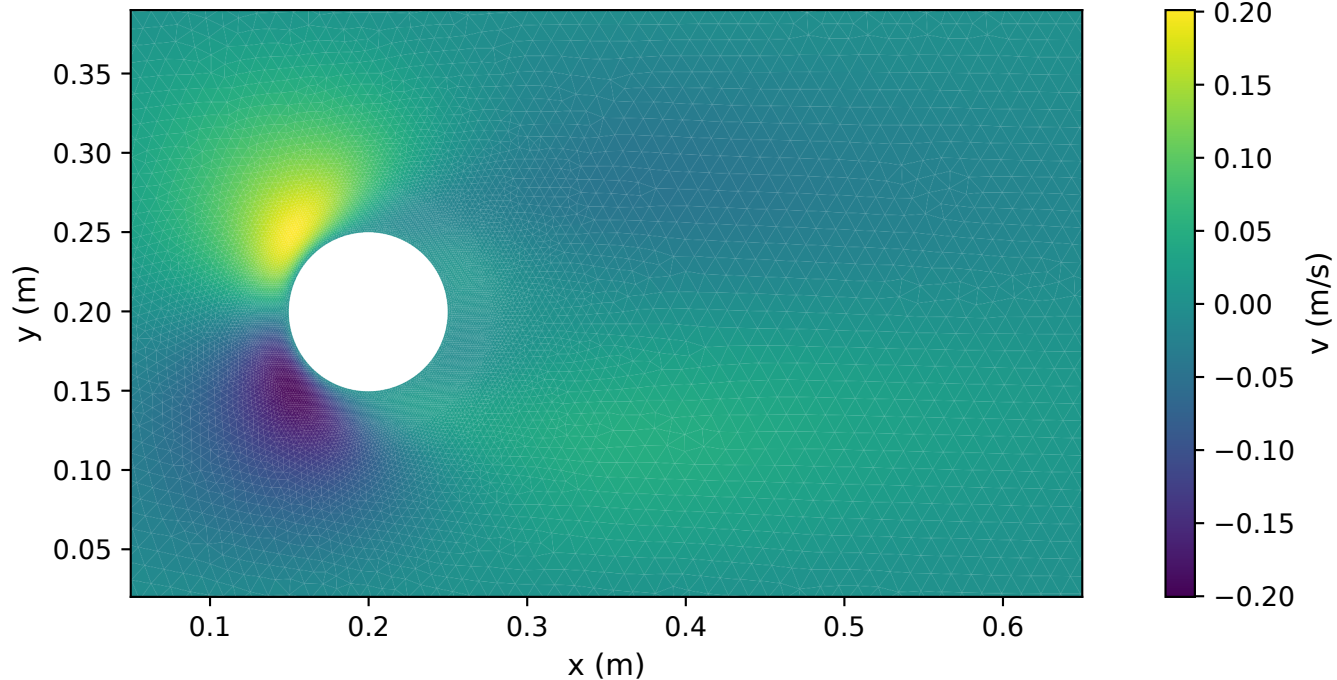
cylinder: 32274 cells



cylinder: 32274 cells



cylinder: 32274 cells



Actual body-fitted mesh

